

Multi-Photon Events

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1 Fidelity in Case of Loss of Photons

Here, we study how the use of a Poisson distribution for the incoming target signal photon number degrades the measured value of the postselected fidelity. We consider events with more than one incoming photon in the *target* pulse when aiming to create a Bell state. An output state with more than one target photon will be a multi-photon GHZ state. If exactly one target photon is detected, then taking the partial trace over the undetected photons yields a maximally mixed two-photon state. For simplicity, we ignore other possible sources of infidelity for these maximally mixed states. Hence, this mixed state has a fidelity of $\frac{1}{2}$ with the desired Bell state. Let F_n denote the fidelity if there are n incoming target photons. Hence

$$F_2 = F_3 = F_4 = \dots = \frac{1}{2}. \quad (1)$$

2 Probability of Loss of Photons

We base our model on two central assumptions.

- (i) There is a Poisson distribution for the number n of incoming target photons.
- (ii) The probabilities of detecting different incoming target photons are uncorrelated.

Let μ denote the mean number of incoming target photons. According to assumption (i), the probability of having n incoming target photons is

$$P_{\text{in}}(n) = \frac{\mu^n}{n!} e^{-\mu}. \quad (2)$$

Let η denote the probability that one incoming target photon is eventually detected. According to assumption (ii), the conditional probability of detecting $k \in \mathbb{Z}_0^+$ target photons conditioned upon the presence of $n \in \mathbb{Z}_0^+$ incoming target photons is the binomial distribution

$$P_{\text{det}}(k|n) = \binom{n}{k} \eta^k (1 - \eta)^{n-k} \times \begin{cases} 1, & k \leq n \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Hence, the total probability of detecting $k \in \mathbb{Z}_0^+$ target photons is

$$\begin{aligned} P_{\text{det}}(k) &= \sum_{n=0}^{\infty} P_{\text{det}}(k|n)P_{\text{in}}(n) = \sum_{n=k}^{\infty} \binom{n}{k} \eta^k (1-\eta)^{n-k} \frac{\mu^n}{n!} e^{-\mu} \\ &= \frac{(\eta\mu)^k}{k!} e^{-\mu} \underbrace{\sum_{n=k}^{\infty} \frac{[(1-\eta)\mu]^{n-k}}{(n-k)!}}_{=e^{(1-\eta)\mu}} = \frac{(\eta\mu)^k}{k!} e^{-\eta\mu}. \end{aligned} \quad (4)$$

This is a Poisson distribution for the number $k \in \mathbb{Z}_0^+$ of detected target photons, which is plausible. Here, $\eta\mu$ is the mean number of detected photons.

For later use, we note that we can use the same calculation with η replaced by $1-\eta$ to show that the total probability of losing $\ell \in \mathbb{Z}_0^+$ target photons is a Poisson distribution

$$P_{\text{loss}}(\ell) = \frac{\lambda^\ell}{\ell!} e^{-\lambda}, \quad (5)$$

where the mean number of lost target photons is $\lambda = (1-\eta)\mu$.

The conditional probability of having $n \in \mathbb{Z}_0^+$ incoming target photons conditioned (i.e. postselected) upon the detection of $k \in \mathbb{Z}_0^+$ target photons is given by Bayes' theorem

$$\begin{aligned} P_{\text{in}}(n|k) &= \frac{P_{\text{in}}(n)}{P_{\text{det}}(k)} P_{\text{det}}(k|n) = \frac{\mu^n}{n!} e^{-\mu} \frac{k!}{(\eta\mu)^k} e^{\eta\mu} \times \begin{cases} \binom{n}{k} \eta^k (1-\eta)^{n-k}, & k \leq n \\ 0, & \text{otherwise} \end{cases} \\ &= \frac{[(1-\eta)\mu]^{n-k}}{(n-k)!} e^{-(1-\eta)\mu} \times \begin{cases} 1, & k \leq n \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (6)$$

To interpret this result, we note that it agrees with $P_{\text{loss}}(\ell)$ for $\ell = n - k \in \mathbb{Z}_0^+$. In other words, the probability of losing ℓ target photons is independent of (i.e. uncorrelated with) the number k of detected target photons.

This result is plausible but we emphasize that it relies on the combination of our two initial assumptions. For example, if we instead assumed a fixed number n of incoming target photons, then postselected upon detected of $k \leq n$ target photons, the number of lost target photons would be fixed to $\ell = n - k$.

3 Averaging

Postselection upon $k = 1$ detected target photons yields the postselected fidelity as the weighted average

$$\begin{aligned} F_{\text{post}} &= \sum_{n=1}^{\infty} F_n P_{\text{in}}(n|1) \stackrel{(1)}{=} F_1 P_{\text{in}}(1|1) + F_2 \underbrace{\sum_{n=2}^{\infty} P_{\text{in}}(n|1)}_{=1-P_{\text{in}}(1|1)} = F_2 + (F_1 - F_2) P_{\text{in}}(1|1) \\ &\stackrel{(6)}{=} F_2 + (F_1 - F_2) e^{-(1-\eta)\mu} \stackrel{(1)}{=} \frac{1}{2} + \left(F_1 - \frac{1}{2}\right) e^{-(1-\eta)\mu}. \end{aligned} \quad (7)$$

Note that, apart from F_1 and F_2 , the result F_{post} depends only on the mean number of lost photons $\lambda = (1-\eta)\mu$. This is quite plausible because it is the loss of photons which leaks information about the $n+1$ photon GHZ state into the environment and taking the partial trace over this environment is what deteriorates F_{post} .

3.1 Free-Space Rydberg Gate

We consider the experiment in Ref. [1] in which the non-unity visibility of single qubits $V_c = 0.66(2)\%$ and $V_t = 0.75(14)\%$ yield $F_1 \leq (1 + V_c)(1 + V_t)/4 + (1 - V_t)/8 = 75.8(6.1)\%$. We combine this with an average number of incoming target photons of $n = 0.50$ (impinging onto the atomic ensemble), an average efficiency of $\eta_{\text{atom}} = 3.0\%$ (of the atomic ensemble), and a probability η_{det} of collecting and detecting a photon (which was transmitted through the atomic ensemble). η_{det} consists of a factor of 0.50 from the quantum efficiency of the detectors [1], a factor between 0.45 [2] and 0.50 [3] for the efficiency for collecting the transmitted light into a single-mode fiber, and of a factor representing the transmission through the second interferometer and the subsequent coupling into a single-mode fiber. It follows from $\eta_{\text{atom}} = 3.0\%$ that $\eta = \eta_{\text{atom}} \times \eta_{\text{det}} \ll 1$ so that $1 - \eta \approx 1$. Hence, the exact value of η_{det} is of little relevance. It also means that taking the average efficiency of the gate instead of the average efficiency of the target light has little effect. The experiment is characterized by

$$F_1 \leq 75.8(6.1)\%, \quad n = 0.50, \quad \eta \ll 1. \quad (8)$$

Inserting this into Eq. (7) yields

$$F_{\text{post}} \leq 65.6(3.7)\%. \quad (9)$$

This agrees within the uncertainties with the measured $F_{\text{post}} = 63.7(4.5)\%$.

3.2 Cavity Rydberg Gate

Alternatively, we consider our recent experiment in a cavity. The efficiency for the target light only is approximated as 100% if the target light is in the bypass rail. If the target photon is in the cavity rail, then this target efficiency is $|\mathcal{R}|^2 = 90(1)\%$ or $|\mathcal{R}_b|^2 = 43(3)\%$ depending on whether a control excitation was stored. To determine the weights with which these events occur, we have to take into account that our Bell-state data are postselected upon the detection of a control and a target photon. The efficiencies are $\eta_{\text{CPHASE}} = (\eta_f, \eta_f|\mathcal{R}|^2, \eta_{\text{sr}}, \eta_{\text{sr},t}|\mathcal{R}_b|^2) = (65\%, 59\%, 39\%, 17\%)$ conditioned upon the input state being one of the CPHASE basis states. We assume that the probability of each incoming CPHASE basis state is $1/4$ because we consider a Bell state measurement. Hence, the total efficiency (in the sense of a total probability) is the arithmetic mean of the above four numbers $\eta_{\text{avg}} = 45\%$. The conditional probabilities that one of these input states was present conditioned upon a two-fold coincidence detection are according to Bayes' law

$$p_{\text{conditional}} = \frac{1/4}{\eta_{\text{avg}}} \eta_{\text{CPHASE}} = (36.1\%, 32.8\%, 21.7\%, 9.4\%). \quad (10)$$

Hence, the postselected probability that the target photon was in the bypass rail is $36.1\% + 21.7\% = 57.8\%$, the postselected probability that the target photon was reflected with coefficient $|\mathcal{R}|^2$ ($|\mathcal{R}_b|^2$) is 32.8% (9.4%). Using these postselected probabilities as weights, we obtain

$$\eta = (57.8\% + 32.8\%|\mathcal{R}|^2 + 9.4\%|\mathcal{R}_b|^2) \times 0.85 \times 0.90 = 69.9\%. \quad (11)$$

We combine this with $n = \bar{n}_t = 0.13$ and assume $F_1 = 1$. We obtain

$$F_{\text{post}} = 98.1\%. \quad (12)$$

For comparison, if we instead use $\frac{1}{4}$ as the weights of the CPHASE basis states, we obtain $\eta \approx \frac{1}{4}(2 + |R|^2 + |R_b|^2) \times 0.85 \times 0.90 = 0.637$. For $F_1 = 1$, this yields $F_{\text{post}} = 97.7\%$. Hence, the exact weights have a moderate effect onto η and little effect onto F_{post} .

References

- [1] D. Tiarks, S. Schmidt-Eberle, T. Stolz, G. Rempe, and S. Dürr, A photon-photon quantum gate based on Rydberg interactions, *Nat. Phys.* **15**, 124–126 (2019).
- [2] S. Schmidt-Eberle, T. Stolz, G. Rempe, and S. Dürr, Dark-time decay of the retrieval efficiency of light stored as a Rydberg excitation in a noninteracting ultracold gas, *Phys. Rev. A* **101**, 013421 (2020).
- [3] D. Tiarks, S. Schmidt, G. Rempe, and S. Dürr, Optical π phase shift created with a single-photon pulse, *Sci. Adv.* **2**, e1600036 (2016).